

Task 1: To Detect Variable References, Expand Values, Handle Undefined Variables, Update Tokens, Support Nested Variables, Test Expansion Logic

```
shreyas@Grigori:~$ mkdir -p ~/skill_session_8
shreyas@Grigori:~$ cd ~/skill_session_8
shreyas@Grigori:~/skill_session_8$ nano variable_expansion.c
shreyas@Grigori:~/skill_session_8$ gcc variable_expansion.c -o variable_expansion
shreyas@Grigori:~/skill_session_8$ ./variable_expansion
/home/shreyas
shreyas
/usr/local/sbin:/usr/local/bin:/usr/sbin:/usr/bin:/sbin:/bin:/usr/games:/usr/local/games:/usr/lib/wsl/lib:/mnt/c/Python314/Scripts:/mnt/c/Python314:/mnt/c/Program Files/Common Files/Oracle/Java/javapath:/mnt/c/Program Files (x86)/Common Files/Oracle/Java/javapath:/mnt/c/Program Files (x86)/Common Files/Oracle/Java/javapath:/mnt/c/Windows/system32:/mnt/c/Windows:/mnt/c/Windows/System32/Wbem:/mnt/c/Windows/System32/WindowsPowerShell/v1.0:/mnt/c/Windows/System32/OpenSSH:/mnt/c/Program Files (x86)/NVIDIA Corporation/PhysX/Common:/mnt/c/Users/shrey/Downloads/jdk-25_windows-x64_bin/jdk-25/bin:/mnt/c/Program Files/NVIDIA Corporation/NVIDIA App/NvDLISR:/mnt/c/Program Files/Git/cmd:/mnt/c/Program Files/nodejs:/mnt/c/ProgramData/chocolatey/bin:/mnt/c/Program Files/PuTTY:/mnt/c/Program Files/MySQL/MySQL Shell 8.0/bin:/mnt/c/Users/shrey/.local/bin:/mnt/c/Users/shrey/Downloads/flutter_windows_3.35.6-stable/flutter/bin:/mnt/c/Users/shrey/AppData/Local/Programs/Python/Python313/Scripts:/mnt/c/Users/shrey/AppData/Local/Programs/Python/Python313:/mnt/c/Users/shrey/AppData/Local/Programs/Python/Launcher:/mnt/c/Users/shrey/AppData/Local/Microsoft/WindowsApps:/mnt/c/wsys64/ucrt64/bin:/mnt/c/Program Files/JetBrains/CLion 2025.2/bin:/mnt/c/Users/shrey/AppData/Local/Programs/Microsoft VS Code/bin:/mnt/c/Program Files/JetBrains/IntelliJ IDEA Community Edition 2025.2.3/bin:/mnt/c/Users/shrey/AppData/Local/JetBrains/Toolbox/scripts:/mnt/c/Program Files/JetBrains/WebStorm 2025.3.2/bin:/mnt/c/Users/shrey/AppData/Local/Programs/Antigravity/bin:/mnt/c/Program Files/JetBrains/PyCharm 2025.2.4/bin:/mnt/c/Users/shrey/AppData/Roaming/npm:/mnt/c/Users/shrey/AppData/Local/Python/bin:/mnt/c/Users/shrey/AppData/Local/GitHubDesktop/bin:/mnt/c/Users/shrey/AppData/Local/Programs/Antigravity IDE/bin:/mnt/c/Users/shrey/Downloads/pypy3.12-v8.8.0-win64:/snap/bin
Undefined variable: UNDEFINED
shreyas@Grigori:~/skill_session_8$ export NAME="Shreyas"
shreyas@Grigori:~/skill_session_8$ export COURSE="AI_DS"
shreyas@Grigori:~/skill_session_8$ echo $NAME
echo $COURSE
Shreyas
AI_DS
```

```
#include <stdio.h>
#include <stdlib.h>
#include <string.h>

void expand_variable(const char *input) {
    char variable[100];

    if (input[0] != '$') {
        printf("%s\n", input);
        return;
    }

    strcpy(variable, input + 1);

    char *value = getenv(variable);

    if (value != NULL) {
        printf("%s\n", value);
    } else {
        printf("Undefined variable: %s\n", variable);
    }
}

int main() {
    expand_variable("$HOME");
    expand_variable("$USER");
}
```

```

    }
}

int main() {
    expand_variable("$HOME");
    expand_variable("$USER");
    expand_variable("$PATH");
    expand_variable("$UNDEFINED");

    return 0;
}

```

Task 2: To Identify Built-ins, Create Dispatch Table, Execute In-Process Commands, Handle Invalid Commands, Maintain State, Test Dispatch Logic

```

#include <stdio.h>
#include <string.h>
#include <stdlib.h>
#include <unistd.h>

void cmd_pwd() {
    char cwd[1024];

    if (getcwd(cwd, sizeof(cwd)) != NULL)
        printf("%s\n", cwd);
}

void cmd_echo(char *arg) {
    printf("%s\n", arg);
}

void cmd_cd(char *path) {
    if (chdir(path) != 0)
        perror("cd");
}

void cmd_export(char *arg) {
    char *equal = strchr(arg, '=');

    if (equal != NULL) {
        *equal = '\0';
    }
}

```

```
| while (1) {  
    printf("shell> ");  
    scanf("%99s", command);  
  
    if (strcmp(command, "exit") == 0)  
        break;  
  
    if (strcmp(command, "pwd") == 0) {  
        strcpy(argument, "");  
    } else {  
        scanf("%99s", argument);  
    }  
  
    int found = 0;  
    for (int i = 0; i < 4; i++) {  
        if (strcmp(command, builtins[i].name) == 0) {  
            builtins[i].function(argument);  
            found = 1;  
            break;  
        }  
    }  
}
```

```
shreyas@Grigori:~/skill_session_8$ nano variable_expansion.c
shreyas@Grigori:~/skill_session_8$ nano builtins.c
shreyas@Grigori:~/skill_session_8$ gcc builtins.c -o builtins
shreyas@Grigori:~/skill_session_8$ ./builtins
shell> pwd
/home/shreyas/skill_session_8
shell> echo hello
hello
shell> cd /tmp
shell> export NAME=Shreyas
shell> abc
exit
Invalid command: abc
shell> echo $NAME
$NAME
shell> ./builtins
^C
shreyas@Grigori:~/skill_session_8$ ./builtins
shell> pwd
/home/shreyas/skill_session_8
shell> ls
```

```
void cmd_export(char *arg) {
    char *equal = strchr(arg, '=');

    if (equal != NULL) {
        *equal = '\0';
        setenv(arg, equal + 1, 1);
    }
}

typedef void (*builtin_function)(char *);

typedef struct {
    char *name;
    builtin_function function;
} Builtin;

void pwd_wrapper(char *arg) {
    cmd_pwd();
}

void echo_wrapper(char *arg) {
    cmd_echo(arg);
}
```